

A Complete Geometric and Algebraic Analysis of a Three-Dimensional Jacobian Conjecture Counterexample

From Polynomial Maps to Rational Normal Curves via Symmetry Reduction

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2026-07-28

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1 Abstract

We present a complete algebraic and geometric dissection of the polynomial map recently proposed as a counterexample to the three-dimensional Jacobian Conjecture. The map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ has constant Jacobian determinant -2 but admits three distinct preimages of the point $(-\frac{1}{4}, 0, 0)$, hence fails to be a polynomial automorphism.

Our analysis proceeds in three layers. First, elementary elimination produces a closed-form expression for the entire fiber $F^{-1}(c, 0, 0)$, revealing a central isolated point plus a $\mathbb{Z}_2$ -paired orbit. Second, projective compactification identifies the non-zero solution branch as a rational normal cubic curve $C \cong \mathbb{P}^1$, realized as the intersection of a quadric cone and a cubic surface in $\mathbb{P}^3$ (in the affine chart $W \neq 0$; the full intersection also contains a line at infinity). Third, the $\mathbb{Z}_2$ involution and $\mathbb{C}^*$ scaling symmetry are shown to govern the full fiber evolution, with $c = 0$ marking the unique degeneration where the non-zero branch collapses to the point at infinity.

We further extract a generalized template $\mathcal{C}_{\alpha, \beta} : XY = \alpha, X^2Z = \beta$, which generates an entire family of counterexamples parametrized by (α, β, c_Z) . The construction reveals that the counterexample is not an isolated coincidence but a manifestation of a $\mathbb{G}_m$ -homogeneous space with a natural $\mathbb{Z}_2$ -quotient. All numerical constants $(2, 3, 26, -1/4)$ are shown to be mere coordinate normalizations that do not affect the projective isomorphism class.

2 1. Introduction

The Jacobian Conjecture, one of the most celebrated open problems in algebraic geometry, asserts that every polynomial map $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with nowhere-vanishing constant Jacobian determinant is a polynomial automorphism. Despite decades of effort, the conjecture remains open for $n \geq 3$.

In 2026, Levent Alpöge and Claude Fable 5 constructed a remarkable three-dimensional polynomial map with constant Jacobian determinant -2 that nevertheless fails to be injective—a single target point has three distinct preimages. This immediately disproves the Jacobian Conjecture in dimension three.

While the original discovery relied on heavy computational algebra and resultant elimination, the present work offers a **complete geometric unpacking** of this counterexample. We show that:

1. The non-zero solution branch is **not** an accidental algebraic curve but a rational normal cubic $C \cong \mathbb{P}^1$;
2. The $\mathbb{Z}_2$ symmetry $(x, y, z) \mapsto (-x, -y, z)$ pairs the two non-zero solutions;
3. The parameter c (the target value of F_1) controls a **continuous fiber evolution**, with $c = 0$ being the unique degeneration point;
4. The entire structure generalizes to a **parametric family** $\mathcal{C}_{\alpha, \beta}$.

Our approach is entirely elementary, requiring only polynomial elimination, elementary projective geometry, and basic group action analysis. It makes the counterexample accessible to a broad mathematical audience and reveals its underlying beauty.

3 2. The Polynomial Map and Basic Properties

3.1 2.1 Definition

Consider the polynomial map $F = (F_1, F_2, F_3) : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ defined by

$$\begin{aligned}
F_1(x, y, z) &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\
F_2(x, y, z) &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\
F_3(x, y, z) &= 2x - 3x^2 y - x^3 z.
\end{aligned} \tag{1}$$

3.2 2.2 The Jacobian Determinant

A direct computation gives

$$\det J_F \equiv -2. \tag{2}$$

Thus F is locally invertible everywhere (an étale map). The constancy of the Jacobian is precisely the hypothesis of the Jacobian Conjecture.

3.3 2.3 Central Symmetry

Define the involution

$$\sigma : (x, y, z) \mapsto (-x, -y, z). \tag{3}$$

The components transform as

$$\sigma(F_1) = F_1, \quad \sigma(F_2) = -F_2, \quad \sigma(F_3) = -F_3. \tag{4}$$

Consequently, the solution set of $F_2 = F_3 = 0$ is invariant under σ : non-zero solutions come in pairs (x, y, z) and $(-x, -y, z)$, while the center $(0, 0, z)$ is a fixed point.

4 3. Algebraic Solution of the Fiber

4.1 3.1 Change of Variables

Introduce

$$u = 1 + xy, \quad A = u^2 z + y^2(1 + 3u). \quad (5)$$

The map becomes

$$\boxed{F_1 = uA, \quad F_2 = y + 3xA, \quad F_3 = x(5 - 3u - x^2 z)}. \quad (6)$$

4.2 3.2 Solving $F_2 = F_3 = 0$

Case 1: $x = 0$.

Substitution gives $y = 0$, and $F_1 = z$. Thus the central solution is

$$P_0(c) = (0, 0, c). \quad (7)$$

Case 2: $x \neq 0$.

From $F_3 = 0$,

$$z = \frac{5 - 3u}{x^2}. \quad (8)$$

Substituting into A and using $y = (u - 1)/x$ yields the identity

$$x^2 A = u + 1. \quad (9)$$

From $F_2 = 0$,

$$y + 3xA = 0 \implies (u - 1) + 3(u + 1) = 0 \implies u = -\frac{1}{2}. \quad (10)$$

Thus $xy = -3/2$ is an invariant. From $F_1 = c$,

$$uA = c \implies -\frac{1}{2} \cdot \frac{1/2}{x^2} = c \implies x^2 = -\frac{1}{4c}. \quad (11)$$

Therefore,

$$\boxed{F^{-1}(c, 0, 0) = \{(0, 0, c)\} \cup \left\{ \left(\pm \frac{1}{2\sqrt{-c}}, \mp 3\sqrt{-c}, -26c \right) \right\}.} \quad (12)$$

4.3 3.3 The Triple Fiber at $c = -1/4$

Setting $c = -1/4$ gives $x^2 = 1$, yielding

$$P_1 = \left(1, -\frac{3}{2}, \frac{13}{2}\right), \quad P_2 = \left(-1, \frac{3}{2}, \frac{13}{2}\right). \quad (13)$$

Together with $P_0 = (0, 0, -1/4)$, the fiber contains exactly three distinct points.

5 4. Projective Compactification

5.1 4.1 Normalized Coordinates

Eliminate the cluttered coefficients by setting

$$X = -2x, \quad Y = -\frac{y}{3}, \quad Z = \frac{z}{26}. \quad (14)$$

On the non-zero branch, $xy = -3/2$ and $x^2z = 13/2$ become

$$XY = -1, \quad X^2Z = 1. \quad (15)$$

5.2 4.2 Projective Closure and the Intersection Curve

Embed into $\mathbb{P}^3$ with homogeneous coordinates $[X : Y : Z : W]$. The projective closures are:

$$Q : \quad XY + W^2 = 0, \quad S : \quad X^2Z - W^3 = 0. \quad (16)$$

Remark (full intersection structure). On the plane at infinity $W = 0$, $Q \cap S$ contains an additional line

$$L = \{[0 : Y : Z : 0]\} \subset \mathbb{P}^3,$$

since $X = 0, W = 0$ makes both Q and S vanish identically. Thus as a set,

$$Q \cap S = C \cup L,$$

where C is the rational normal cubic of interest. The line L lies entirely at infinity ($W = 0$) and does not participate in the affine fiber analysis. Hence we focus on the component C in the affine chart $W \neq 0$.

Proposition 1. The portion of $Q \cap S$ in the affine chart $W \neq 0$ is a rational normal cubic curve C with homogeneous parameterization

$$\boxed{C : \quad [X : Y : Z : W] = [t^3 : -t : 1 : t^2], \quad t \in \mathbb{P}^1.} \quad (17)$$

In the affine chart $W = 1$,

$$X = t, \quad Y = -\frac{1}{t}, \quad Z = \frac{1}{t^2}. \quad (18)$$

Proof. Direct substitution verifies both equations. $\square$

5.3 4.3 Hyperplane Sections

On C , the condition $F_1 = c$ becomes $Z = -c$, i.e., $t^2 = -1/c$. Hence for $c \neq 0$, the hyperplane

$$H_c : Z + cW = 0 \tag{19}$$

intersects C in exactly two points $t = \pm\sqrt{-1/c}$, which are paired by the $\mathbb{Z}_2$ involution.

Theorem 1 (Complete Fiber Structure). For every $c \in \mathbb{C}^*$,

$$\boxed{\text{Fiber}(c) = (C \cap H_c) \sqcup \{P_c\},} \tag{20}$$

where

$$P_c = \left[0 : 0 : \frac{c}{26} : 1 \right] \tag{21}$$

is the isolated central point. The denominator 26 ensures $z = 26Z = c$ in the affine chart.

5.4 4.4 The Degenerate Fiber $c = 0$

When $c = 0$, the equation $t^2 = \infty$ has no finite solution. The unique point on C satisfying $Z = 0$ is $[1 : 0 : 0 : 0]$ (corresponding to $t = \infty$), but this point lies at infinity ($W = 0$) and is not in the affine chart. Hence in the affine space,

$$C \cap H_0 = \emptyset. \tag{22}$$

The non-zero branch collapses to the point at infinity $[1 : 0 : 0 : 0]$, leaving only the isolated central point $P_0 = [0 : 0 : 0 : 1]$ (corresponding to the affine origin $(0, 0, 0)$).

5.5 4.5 The Points at Infinity

Two special homogeneous points merit attention:

t	Point	c limit	Role
0	$[0 : 0 : 1 : 0]$	$c \rightarrow \infty$	Branch point of ι
∞	$[1 : 0 : 0 : 0]$	$c \rightarrow 0$	Collapse point of non-zero branch

Both lie in $W = 0$ (the plane at infinity) and are the fixed points of the $\mathbb{Z}_2$ involution.

6 5. Group Actions and Symmetries

6.1 5.1 The $\mathbb{Z}_2$ Involution

The involution

$$\iota : [X : Y : Z : W] \mapsto [-X : -Y : Z : W] \quad (23)$$

acts on C as $t \mapsto -t$. Its fixed points are exactly $t = 0$ and $t = \infty$, the two points at infinity.

The quotient $C/\iota \cong \mathbb{P}^1$ is parametrized by $u = t^2$.

6.2 5.2 The $\mathbb{C}^*$ Scaling Symmetry

In homogeneous coordinates,

$$[X : Y : Z : W] \mapsto [\lambda^3 X : \lambda Y : Z : \lambda^2 W], \quad \lambda \in \mathbb{C}^*. \quad (24)$$

Pulling back to the original (x, y, z) coordinates using (14) gives

$$\boxed{(x, y, z) \mapsto (\lambda x, \lambda^{-1}y, \lambda^{-2}z).} \quad (25)$$

Under this scaling, the map components transform equivariantly:

$$\begin{aligned} F_1(\lambda x, \lambda^{-1}y, \lambda^{-2}z) &= \lambda^{-2}F_1(x, y, z), \\ F_2(\cdot) &= \lambda^{-1}F_2(x, y, z), \quad F_3(\cdot) = \lambda F_3(x, y, z). \end{aligned} \quad (26)$$

6.3 5.3 Fiber Evolution Phase Diagram

c range	$C \cap H_c$	Central point	Total
$c \in \mathbb{C}^*$	2 points ($\mathbb{Z}_2$ orbit)	1 point	3
$c = 0$	$\emptyset$ (collapsed to $[1 : 0 : 0 : 0]$)	1 point	1

- **Generic fiber:** The non-zero branch is $\mathbb{G}_m \cong \mathbb{C}^*$, parametrized by t or X ;
- **Group action:** $\mathbb{Z}_2 = \{\pm 1\}$ acts freely via $t \mapsto -t$, with quotient $\mathbb{G}_m/\mathbb{Z}_2 \cong \mathbb{G}_m$;
- **Base space:** Spanned by the parameter c (the value of F_1), i.e., $\text{Spec } \mathbb{C}[c]$;
- **Compactification:** $\mathbb{P}^1$ compactification discards $t = 0$ (corresponding to $c \rightarrow \infty$, point $[0 : 0 : 1 : 0]$) and $t = \infty$ (corresponding to $c = 0$, point $[1 : 0 : 0 : 0]$);
- **Degeneration:** At $c = 0$, the non-zero fiber collapses to $[1 : 0 : 0 : 0]$, leaving only the isolated central point; as $c \rightarrow \infty$, the paired orbit escapes to $[0 : 0 : 1 : 0]$.

7 6. Generalized Template: A Family of Counterexamples

The structure revealed above can be abstracted. Define

$$\mathcal{C}_{\alpha, \beta} : \quad XY = \alpha, \quad X^2Z = \beta, \quad \alpha, \beta \in \mathbb{C}^*. \quad (27)$$

(In this section, c denotes the Z -coordinate value, which differs by a sign from the c in $F_1 = c$; the reader is advised to note this distinction.)

Then:

1. $\mathcal{C}_{\alpha,\beta} \cong \mathbb{G}_m$, parametrized by $X = t$;
2. The $\mathbb{Z}_2$ involution $\iota : (X, Y, Z) \mapsto (-X, -Y, Z)$ sends $t \mapsto -t$;
3. The hyperplane section $Z = c_Z$ gives $t^2 = \beta/c_Z$, exactly two points;
4. The $\mathbb{C}^*$ scaling $(X, Y, Z) \mapsto (\lambda X, \lambda^{-1}Y, \lambda^{-2}Z)$ preserves $\mathcal{C}_{\alpha,\beta}$.

Thus the construction generates a **parametric family of counterexamples** indexed by (α, β, c_Z) . The original Alpöge–Fable 5 counterexample corresponds to $(\alpha, \beta) = (-1, 1)$ with F_1 target value $-1/4$ (i.e., $Z = 1/4$).

8 7. Conclusion

We have presented a complete algebraic and geometric analysis of the three-dimensional Jacobian Conjecture counterexample. The main results are:

1. **Explicit fiber formula:**

$$F^{-1}(c, 0, 0) = \{(0, 0, c)\} \cup \left\{ \left(\pm \frac{1}{2\sqrt{-c}}, \mp 3\sqrt{-c}, -26c \right) \right\}.$$

2. **Projective geometric meaning:** The non-zero branch is a rational normal cubic curve $C \cong \mathbb{P}^1$, the component of $Q \cap S$ in the affine chart $W \neq 0$;
3. **Symmetry structure:** The $\mathbb{Z}_2$ involution pairs the two non-zero solutions; the $\mathbb{C}^*$ scaling gives continuous deformation; $c = 0$ is the unique degeneration point where the non-zero branch collapses to infinity;
4. **Generalized template:** A full family $\mathcal{C}_{\alpha,\beta}$ parametrizes all such constructions, showing that the counterexample is not an isolated coincidence.

The map has constant Jacobian determinant -2 but is not injective, thereby disproving the Jacobian Conjecture in dimension three. The geometric framework presented here makes the counterexample transparent and amenable to further study.

9 References

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